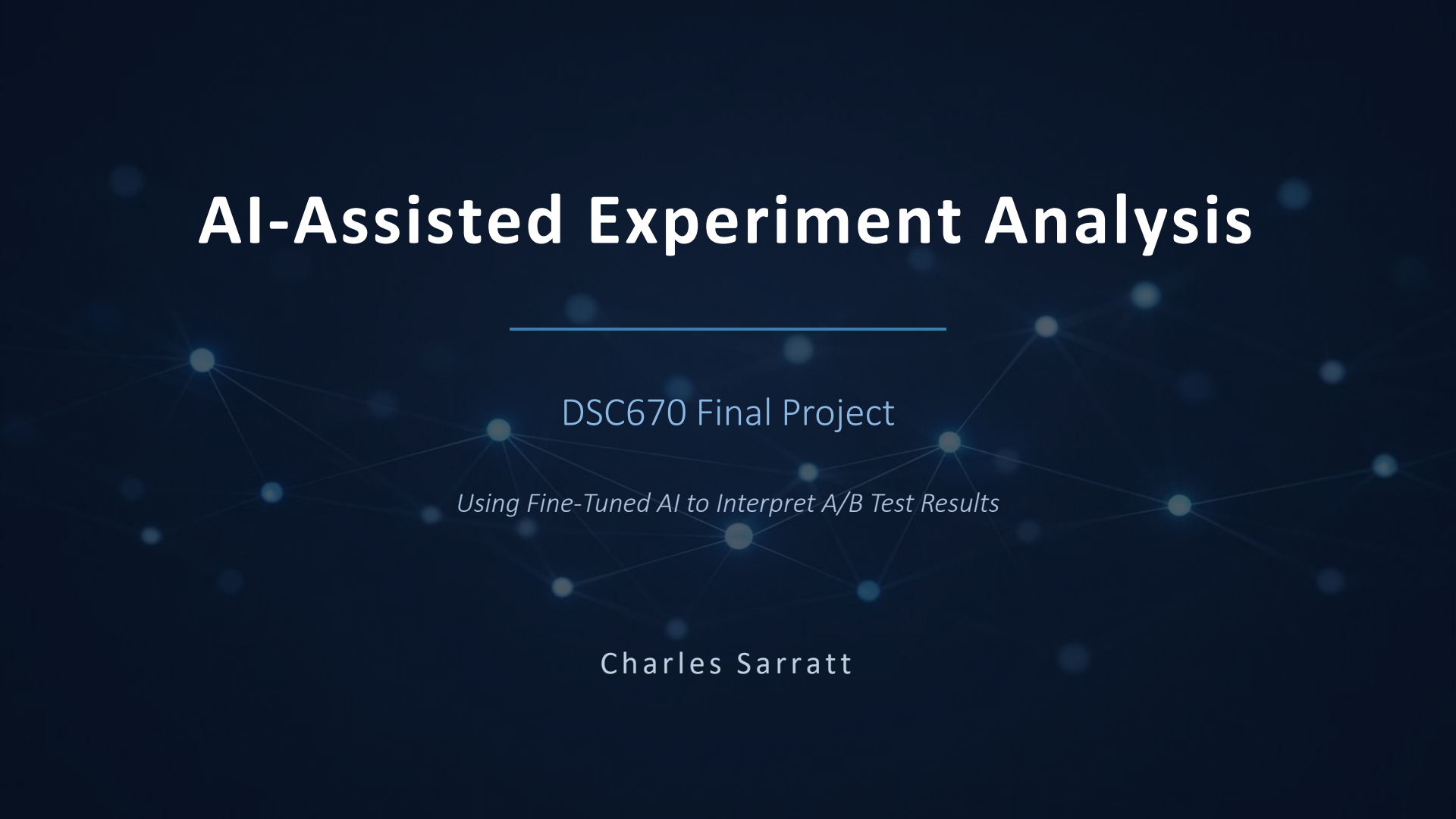


AI-Assisted Experiment Analysis



DSC670 Final Project

Using Fine-Tuned AI to Interpret A/B Test Results

Charles Sarratt

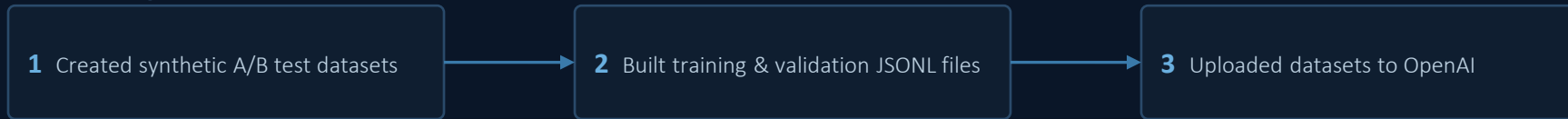
Project Objective

- Built a generative AI-assisted application to interpret A/B test results and produce structured, actionable recommendations
- Focused on evaluating how OpenAI fine-tuning could improve output consistency and response structure
- Designed to support experimentation analysis workflows while keeping human decision-making central to the process
- Compared prompt engineering and fine-tuning approaches for interpreting A/B test results
- Deployed the final application as a Streamlit web application for interactive use

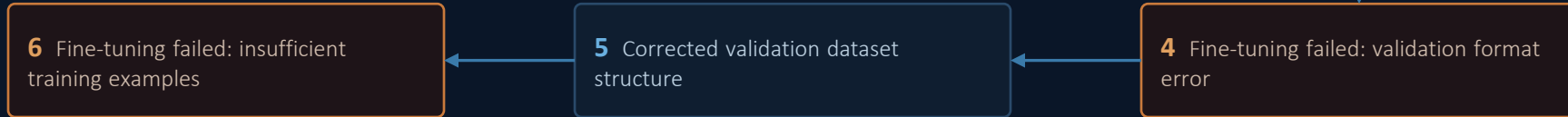


Fine-Tuning Workflow

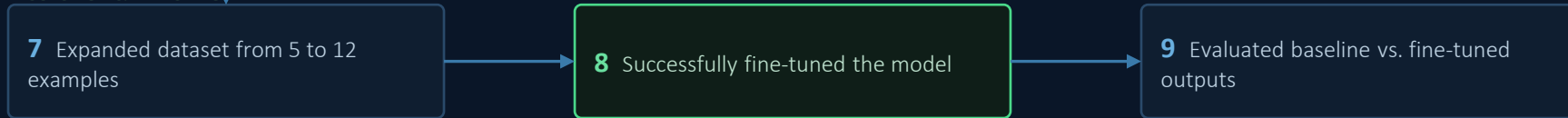
DATA PREPARATION



ITERATION & DEBUGGING



RESOLUTION & EVALUATION



Note: Synthetic data was used because real experimentation data was unavailable or proprietary. Multiple fine-tuning iterations refined the overall implementation process.

Model Evaluation

Baseline Model

- Correctly interpreted statistically significant and non-significant A/B test results
- Produced longer, less structured responses with more variability
- Added explanatory details and assumptions beyond the provided metrics

vs

Fine-Tuned Model

- Produced consistent Summary, Interpretation, and Recommendation sections
- Generated shorter, more controlled and predictable responses
- Avoided unnecessary assumptions outside the provided data

KEY CONCLUSIONS

- Fine-tuning improved output consistency and response structure more than core reasoning capability
- Prompt engineering alone proved highly effective for accurate interpretation of A/B test results
- Fine-tuning added practical value through predictable formatting and controlled, concise outputs

Streamlit Application

Interactive web application for AI-assisted A/B test analysis

AI-Assisted Experiment Analysis

This application uses a fine-tuned OpenAI model to interpret A/B test results and generate a structured recommendation.

Enter A/B Test Metrics

Control Conversion Rate (%)

10.20

- +

Variant Conversion Rate (%)

11.80

- +

Sample Size Per Group

8000

- +

P-value

0.040

- +

Lift (%)

15.70

- +

Analyze Experiment

Input

Input Sent to Model

```
Control Conversion Rate: 10.2%  
Variant Conversion Rate: 11.8%  
Sample Size: 8,000 per group  
P-value: 0.04  
Lift: +15.7%
```

Model Recommendation

Summary

The variant outperformed the control with a higher conversion rate.

Interpretation

The result is statistically significant because the p-value is below 0.05, and the lift indicates a meaningful improvement.

Recommendation

Implement the variant while monitoring post-launch performance.

Output

A/B Test Metric Inputs

Users enter control/test values, sample size, and p-value interactively

Structured AI Output

Summary, Interpretation, and Recommendation sections from fine-tuned model

Practical Implementation

Demonstrates practical prompt engineering and fine-tuning concepts in a deployed application

Human Oversight Required

Synthetic data notice included; AI output augments rather than replaces analyst judgment

The fine-tuned model generates structured experiment analysis from user-provided A/B test metrics in real time

Synthetic Data Risks and Limitations

WHY SYNTHETIC DATA

- Real experimentation datasets are often proprietary, sensitive, or unavailable for academic research
- Synthetic examples enabled controlled testing and repeatable training scenarios

- Synthetic data served as a practical starting point for exploring fine-tuning techniques
- Results demonstrate the viability of the fine-tuning methodology while acknowledging dataset limitations

PROJECT CONCLUSION

- This project should be viewed as an **experimental implementation** exploring fine-tuning methodology, not a production-ready analytics system
- Human oversight remains essential when interpreting experiment outcomes in any applied setting

RISKS & LIMITATIONS

- Synthetic data can inherit assumptions and biases from the system that generated it
- AI-generated training examples may oversimplify the complexity of real-world experimentation data
- Small synthetic datasets limit the model's ability to generalize across diverse scenarios
- Fine-tuned outputs may appear structurally consistent without actually improving reasoning quality
- Real business experimentation environments often contain noisy, incomplete, or conflicting data

Future Improvements

DATA IMPROVEMENTS

- Larger real-world experimentation datasets
- More diverse and noisy business scenarios

APPLICATION ENHANCEMENTS

- Experiment history tracking
- Statistical dashboards and visualizations
- Additional experimentation metrics

MODEL ENHANCEMENTS

- Expanded fine-tuning datasets
- Evaluate additional model variants and prompting strategies
- Confidence scoring for recommendations

DEPLOYMENT CONSIDERATIONS

- Authentication and role-based access
- Integration with experimentation platforms
- Expert validation before production deployment

Final Conclusions

KEY OUTCOMES

- Built a complete fine-tuning workflow from synthetic data generation through model deployment and evaluation
- Developed a working Streamlit application that demonstrates real-time AI-assisted A/B test interpretation
- Fine-tuning improved output consistency and structural formatting compared to baseline prompt engineering alone

MAJOR FINDINGS

- Prompt engineering alone achieved strong baseline results, highlighting its effectiveness for structured output tasks
- Fine-tuning added the most value in formatting consistency and adherence to output structure requirements
- Iterative debugging and refinement of the training data proved essential for improving model performance

FINAL REFLECTION

- AI-generated analysis can support decision-making but requires human oversight to ensure accuracy and context
- Synthetic training data is a useful starting point but introduces inherent limitations on generalizability
- Production deployment would require larger, real-world datasets and rigorous validation against expert analysis

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GitHub Project Repository

github.com/csarratt/portfolio/tree/main/ai-assisted-experiment-analysis